

IOT SECURITY CHALLENGES

INTRODUCTION TO IOT AND SECURITY

What is IoT?

The Internet of Things (IoT) is a network of physical devices connected to the internet that can collect, exchange, and process data without human intervention.

Examples of IoT Devices

- ❖ Smart Home Systems
- ❖ Smart TVs
- ❖ Smart Watches
- ❖ Security Cameras
- ❖ Smart Agriculture Sensors
- ❖ Industrial Automation Systems

Importance of IoT Security

IoT security protects devices, networks, and data from cyberattacks. As billions of devices become connected, ensuring security is critical for protecting privacy and maintaining reliable services.

Major Security Goals

- ❖ Confidentiality
- ❖ Integrity
- ❖ Availability
- ❖ Authentication
- ❖ Privacy Protection

SECURITY CHALLENGE IN IOT DEVICES

- ❖ Major Vulnerabilities
- ❖ Weak Authentication
- ❖ Many IoT devices use factory-default usernames and passwords such as:
 - ❖ admin/admin
 - ❖ root/root
 - ❖ user/password
- ❖ Lack of Firmware Updates
- ❖ Manufacturers often fail to provide regular security patches.

❖ Limited Processing Power

❖ Many devices cannot support advanced security mechanisms because of limited hardware resources.

❖ Poor Encryption

❖ Data transmitted between devices is often not encrypted properly.

❖ Insecure Network Configuration

❖ Open ports and unsecured services create opportunities for attackers.

HOW THE MIRAI ATTACK WORKED

Step-by-Step Attack Process

Step 1: Internet Scanning

- ❖ Mirai continuously scanned the internet searching for vulnerable IoT devices.

Step 2: Credential Guessing

- ❖ The malware attempted to log in using a list of common default usernames and passwords.

Step 3: Device Infection

- ❖ When login was successful, malware was installed on the device.

Step 4: Botnet Creation

- ❖ Thousands of infected devices connected to a command-and-control server.

Step 5: DDoS Attack Launch

- ❖ The attacker instructed all infected devices to send massive traffic to a target server simultaneously.

Result

- ❖ Target servers became overloaded and stopped responding to legitimate users.

IMPACT OF THE ATTACK

- ❖ Major Consequences

- ❖ Service Disruptions

- ❖ Several popular websites became inaccessible.

- ❖ Economic Losses

- ❖ Businesses experienced financial losses due to downtime.

- ❖ Reduced Customer Trust

- ❖ Consumers became concerned about IoT device security.

❖ Increased Security Awareness

❖ Organizations realized the importance of cybersecurity in connected systems.

❖ Services Affected

❖ Twitter

❖ Netflix

❖ Reddit

❖ Spotify

❖ GitHub

❖ PayPal

❖ Millions of users experienced service interruptions.

ROOT CAUSE ANALYSIS

Why Was the Attack Successful?

- ❖ Poor Password Management
- ❖ Most users never changed default credentials.
- ❖ Lack of Security Standards
- ❖ Manufacturers prioritized low cost and convenience over security.
- ❖ Insufficient Monitoring
- ❖ Many organizations lacked systems to detect suspicious network activities.

❖ Absence of Automatic Updates

❖ Devices remained vulnerable for long periods.

❖ User Awareness Issues

❖ Consumers were often unaware of cybersecurity risks.

❖ Security Gap

❖ Security was treated as an optional feature rather than a core requirement

SOLUTION AND COUNTERMEASURES

- ❖ Strengthening IoT Security

- ❖ Strong Authentication

- ❖ Use unique passwords.

- ❖ Implement Multi-Factor Authentication (MFA).

- ❖ Regular Firmware Updates

- ❖ Manufacturers should provide security updates throughout the device lifecycle.

- ❖ Network Segmentation

- ❖ Separate IoT devices from critical business systems.

❖ Encryption

❖ Use secure protocols such as:

❖ HTTPS

❖ TLS

❖ SSL

❖ Security Monitoring

❖ Deploy Intrusion Detection Systems (IDS) and continuous monitoring solutions.

❖ Secure Device Configuration

❖ Disable unused services and close unnecessary network ports.

LESSONS LEARNED AND RECOMMENDATION

- ❖ Lessons Learned

- ❖ Security by Design

- ❖ Security should be incorporated during the product development phase.

- ❖ Continuous Monitoring

- ❖ Organizations must monitor device activities and network traffic.

❖ User Education

❖ Consumers should be trained to:

❖ Change passwords

❖ Update firmware

❖ Follow cybersecurity best practices

❖ Industry Collaboration

❖ Manufacturers, governments, and security experts should work together to establish security standards.

❖ Risk Management

❖ Organizations should regularly assess and manage cybersecurity risks.

CONCLUSION

- ❖ The Mirai Botnet attack demonstrated the serious security challenges associated with Internet of Things devices. Weak authentication methods, poor security configurations, and the absence of regular updates allowed attackers to compromise a large number of connected devices. The attack caused widespread disruption and highlighted the urgent need for stronger IoT security measures.
- ❖ To ensure a secure IoT ecosystem, manufacturers must implement security-by-design principles, provide timely software updates, and adopt strong encryption techniques. Users must also take responsibility by changing default passwords, updating devices regularly, and following cybersecurity best practices.
- ❖ As IoT adoption continues to grow in smart homes, healthcare, agriculture, transportation, and industrial systems, robust security measures will be essential for protecting data, maintaining service availability, and ensuring public trust in connected technologies